DCLAB · OFFICIAL DOCUMENT

DCLab Vision

> From raw data to trusted models, and from models to better decisions.

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1. The vision

DCLab is an AI-powered platform that takes companies from raw data to trusted models, and from models to better decisions.

For data scientists and ML engineers, DCLab is a smart assistant. An AI agent first understands the problem, then designs and runs the right ML pipeline. Before a model leaves DCLab, it is checked: no leakage, an honest score, and the code that rebuilds it. Developers work from their own notebooks and tools, and spend their time on the real problem instead of repetitive work.

For businesses, DCLab goes one step further. Soon every company will run many models, each making its own predictions. But predictions alone are not enough. Companies need help turning them into the best decisions. DCLab will give each company its own decision model, built on its own data, that grows together with the company and its business.

Every business decision workflow starts with a data science and ML team. DCLab supports that team and the decision-makers they serve, as one assistant for both.

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2. Mission

Our daily work, in one sentence:

> **Help every data team build models that are right, help every business turn those models into its best decisions, and keep learning from what happens next.**

The vision says where we are going. The mission says what we do every day to get there. Each of its three parts matches one layer of the product (section 5).

3. The problem we solve

3.1 A prediction is not a decision

A model says what will probably happen. A decision says what we do about it. Between the two sit costs, budgets, capacity, rules and timing. Most tools stop at the prediction and leave the decision to a spreadsheet or to a feeling.

> **Example (from DCLab's own work).** A bank's term-deposit campaign. The model says who will probably say yes. Say the bank can make only 200 calls a day. Who is called first? A customer who will say yes anyway does not need a call; a customer who will never say yes wastes one. The best list is not "the highest scores". That choice is the decision, and the model alone does not make it.

3.2 Many models, one customer

Companies already run several models, and the number grows each year. Each model is built by a different person, at a different time, for a different goal. They are rarely combined.

> **Example.** A telecom company. A churn model says who will leave. A value model says who is worth keeping. An uplift model says who an offer would actually change. The budget is 500 discounts a month. Who gets them? Each model alone gives a different list. The company needs one answer.

3.3 Models that look good but are wrong

The most common failure in applied machine learning is not a weak algorithm. It is a model that uses information it will not have at the moment of the decision (leakage), or that is scored on data it has already seen. Such a model looks excellent in the notebook and fails in use.

> **Example (measured in this repository).** A bank-marketing notebook printed an accuracy of 0.9689. It used the length of the call, which is known only after the call ends. After DCLab's review and fixes, the honest score was ROC-AUC 0.7577 (average precision 0.3454, on 9,043 test rows). Lower, but true.

3.4 Data teams lose their time to repetitive work

Data scientists spend much of their time on the same steps: reading the data, cleaning it, choosing a split, checking for leakage, trying model families, writing the report. These steps matter, but they repeat. The time left for the real question, "what should the business do?", is small.

3.5 Decisions do not learn

Few companies record what they decided, why, and what happened after. Without that record, the next decision cannot be better than the last one, and no model can learn the company's own way of working.

4. Who we serve

Who	What they need	What DCLab gives them
Data scientists and ML engineers	Build correct models fast; avoid leakage and false scores; less repetitive work; work in their own notebook and tools.	An agent that understands the problem, designs and runs the pipeline, checks every model before export, and gives back code that rebuilds the result. Available in the notebook, the chat, the DCLab studio and other tools through MCP.
Decision-makers (managers, heads of operations, risk, marketing)	A clear action, not a score: who, what, how many, why, and what it will bring.	A decision card: the recommended actions, the expected gain with its range, the reasons with every number traced to its source, and what changes if the budget changes.
The company (leadership, owners of the data)	Decisions that improve over time; control over its data; a lasting asset, not a one-off project.	Its own decision model, trained on its own decisions and outcomes, owned by the company, used for training only with its permission, replaced only by a version that measured better.

Why both groups. The decision-maker asks the question. The data team builds the models that answer it. If the models are wrong, the decision is wrong too. So DCLab serves both, on one shared record: the team sees how each number was made; the decision-maker sees what it means and what to do. Like building a house: the owner says what they need, the architect makes it stand, and both read the same plan.

5. What DCLab is: three layers

Layer 1 · Trusted models: the assistant for the data team

- **Understand first.** The agent asks questions, writes down what it heard and where each fact came from (the person, a calculation, or a guess), and writes the prediction contract before looking at any score.
- **Design and run the pipeline.** The split, the leakage audit, features, model families, tuning and an honest evaluation, across many kinds of problem: classification, regression, fraud and rare events, anomalies, forecasting, time to event, uplift, ranking, segmentation, related tables, streams, places and text.
- **Check before export.** No leakage, the final test read once, every score with its uncertainty, and exported code that rebuilds the same final number.
- **Work where the team works.** A Python SDK in the notebook, the DCLab studio, the chat, a developer app, and any tool that speaks MCP.

Layer 2 · Decisions: from predictions to actions

- **One model, one decision.** The operating point chosen from costs and value, a list sized to the team's capacity, and uplift (act only where the action changes the result).
- **Many models, one decision.** Several models' predictions about the same customer, order or case, combined under shared budgets, capacity and rules into one best action.
- **Explained.** Every recommendation says why, with every number traced to the record that made it, and shows what changes if a cost or a budget changes.

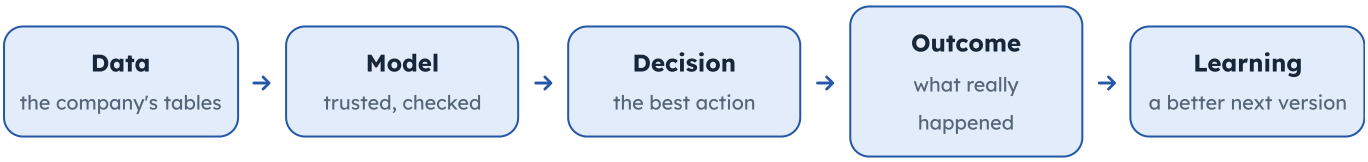
Layer 3 · Learning: the company decision model that grows

- **Record.** Every prediction, decision and outcome is logged, without copying the company's table values into the record.
- **Watch.** Monitoring raises an alert when new data stops looking like the data the model learned from.
- **Improve.** Retraining with champion and challenger: a new version replaces the old one only when it wins on the newest data.
- **Keep learning honestly.** A small, agreed share of cases gets a different action, so the model does not learn only from its own choices.
- **Become the company's own.** Over time, the company gets its own small models, trained on its own decisions and outcomes, with its consent.

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6. The DCLab loop

The three layers form one loop. Each turn of the loop makes the next decision better.



↻ Learning feeds back into the model and the decision.

Like a new employee. On day one, they know the job in general. After a year, they know your customers, your costs and your habits, because they saw what worked and what did not. DCLab's decision model should grow the same way.

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7. Principles

These rules hold in every layer. They are already enforced in DCLab's code, and they do not change for speed or for a better-looking number.

1. **Understand before building.** The problem, the decision and the moment of prediction are written down before any model is trained.
2. **No leakage.** Nothing is used that is unknown at the moment of the decision.
3. **The final test is read once.** Every score is reported with its uncertainty and its limits.
4. **Every number is traced.** Each number in a report links to the run and the record that made it.

5. **The AI suggests; code checks.** Splits, metrics and selection are owned by deterministic code. Rules, validators and gates have the last word.
6. **Promote only what measured better.** A new model, rule or agent is used only after it beat the current one on data it had not seen.
7. **The company's data stays the company's.** Nothing is used to train any model without the company's permission; table values never leave in records.

What we do not promise. We never call a model production-ready from a test score alone. A checked model is one that is free of the errors we can detect; its value in use must be measured in use.

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8. Why DCLab can win

Many tools build models, and many chat assistants write code. DCLab's difference is that every step comes with proof, and every decision is recorded so that it can be learned from. These strengths are built; they still have to be proven with real companies.

- **Evidence behind the advice.** 579 recorded experiments on real public and project data, turned into rules, pitfalls and leakage precedents that the agent cites.
- **Leakage first.** Leakage review, train-only exploration and a single read of the final test are built into the workflow, not left to the user.
- **Traceable results.** Lineage links every reported number to its run; the code export rebuilds it in a clean environment.
- **A record of every decision.** A decision ledger of 45 decision points (28 technical, 17 workflow) records what was decided and why. This record is the raw material of each company's own decision model.
- **Measured, not assumed.** Agents and models are judged on sealed exams that are read once, and promoted only where they beat the current way.
- **Model freedom.** DCLab runs with Claude, with local open models, or with rules alone; its own small models take over a task only where they measured better.

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9. Where we are today

An honest account, as of 1 November 2026.

Area	Status	Details
Problem engine	Built	15 kinds of problem supported, 2 partial (metric change, embedded data); images not yet.
Leakage checks, single read of the final test, code export that rebuilds the number	Built	Checked for every engine kind by `make code-check`.
Agent that understands first (the lead agent)	Partly proven	Second sealed exam (19 written people): understanding 0.93 against 0.82 for Claude alone; 0 invented numbers against 139. It failed its gate on choosing the right route (0.79 against 1.00) and on traps. Fixes are in; a fresh sealed exam is next.

Area	Status	Details
Decision pieces for one model	Built, not used in a company	Operating point from costs and value per case, review list sized to capacity, uplift, the prediction-decision-outcome log, monitoring, retraining with champion and challenger.
Open-source studio (`dclabai`) and the developer app (DCLab Code)	Built	The open core is public on GitHub; DCLab Code is private.
Python SDK for notebooks	Not built	Today: a notebook companion and a notebook review from the command line, and MCP tools.
Many models, one decision	Not built	The heart of the business vision.
Business view for decision-makers	Not built	Today's pages are made for data scientists.
Running inside a real company; company-owned models	Not yet	No real team uses DCLab weekly yet, and no company has sent outcomes back.

The plan to close these gaps is in the companion document, The Plan to Reach the Vision.

10. What success looks like

Proposed measures, to be confirmed with the first real users. Each is measured, not estimated.

Layer	Measure
Trusted models	Data teams using DCLab every week. Real leaks and flaws caught before export, confirmed by the team. Time from raw data to the first trusted model.
Decisions	Measured gain of decisions made with DCLab against the company's previous rule, with its range. Share of promised gains later confirmed by outcomes.
Learning	Companies whose own model beats the general one on new decisions, without losing what it knew before.
The company	Paying customers, and how long they stay.

North-star measure: the measured value of the decisions made with DCLab.

11. Words we use

Word	Meaning in this document
Company decision model	The model that turns many predictions, plus the company's costs, budgets and rules, into one best action, and learns from outcomes. Not the same as the ledger's decision models below.
Decision points, decision models (ledger)	DCLab's own small models that answer its technical and workflow choices, such as a column's role or the next step. Internal tools.
Leakage	Using information that is not known at the moment of the prediction or decision.
Holdout (final test)	Data kept apart and read once, to give an honest final score.
Outcome	What really happened after a decision, often known weeks later.
Uplift	How much an action changes the result for a case, compared with doing nothing.
Champion and challenger	The model in use and the new candidate; the challenger replaces the champion only if it wins on the newest data.
Sealed exam	A test written by someone else and run once, so that its score cannot be tuned.

12. Related documents

- The Plan to Reach the Vision: the gaps and phases 14 to 19.
- `docs/vision/VISION.md`: the vision statement, short form.
- `docs/guides/THREE_MODELS.md`: DCLab's own models and the data each learns from.
- `docs/guides/ROADMAP.md`: the packages built so far, phase by phase.